

SRINIVAS UNIVERSITY, MANGALORE
INSTITUTE OF COMPUTER AND INFORMATION SCIENCE (ICIS)

Question Bank – 2025-2026

Subject: Cyber Security and Privacy Practices

Semester: MCA II Semester

UNIT 1:

1- Mark Questions

CO1 (5)

1. What is Internet Security?
2. Define Undirected Threats.
3. Define Directed Threats.
4. What is malware?
5. What is phishing?

CO2 (5)

1. Explain the role of a firewall.
2. What is email security?
3. How does end-to-end encryption work?
4. Why is MFA important?
5. What is URL filtering?

CO3 (5)

1. You find an unknown USB drive on your desk. What is the safest action you should take?
2. Your computer becomes unusually slow. What is one security-related reason you would check?
3. You accidentally clicked on a suspicious link. What should you do next?
4. You need to share an important file online. What is one way to keep it secure?
5. You receive multiple OTP requests without trying to log in. What action should you take?

7-Mark Questions

CO1 (5)

1. Define Internet Security and threats.
2. Explain symmetric vs asymmetric encryption.
3. Explain basic security practices.
4. Explain end-to-end encryption.
5. Explain Types of cyber threats.

CO2 (5)

1. Explain how a firewall and antivirus work together to enhance system security.
2. Describe the end-to-end (E2E) encryption process with an example.
3. Discuss the importance of using secure browsers in protecting user data.
4. Differentiate between trusted software and open-source software.
5. Explain the concept of data loss prevention (DLP) and its significance.

CO3 (5)

1. A company's firewall is blocking legitimate employee access to a website. How would you identify and resolve the issue?
2. You are asked to improve security in an internet café. What practical measures would you implement and why?
3. Your personal computer shows signs of malware infection. Describe the steps you would take to secure and recover it.
4. An employee receives suspicious email requesting login credentials. How should secure email practices be applied to handle this situation?
5. A small business has recently experienced repeated phishing attacks. Develop a basic cybersecurity protection plan to address this problem.

10 Mark Questions

Application-Based

1. What plan would you make for implementing multi-layered cybersecurity in a small business setup? Include software, hardware, and user-awareness components.
2. Predict what would happen if an organization does not use encryption while transmitting sensitive data over the internet.

UNIT 2: Password Security and Trusted Systems

1-Mark Questions

CO1 (5)

1. What is a password manager?
2. Define Trusted Systems.
3. What is MFA?
4. What is password complexity?
5. List security levels in MLS.

CO2 (5)

1. Explain the importance of strong passwords in ensuring cybersecurity.
2. Explain Read Up / Write Down.
3. What is TTL in passwords?
4. Explain why password reuse should be avoided in cybersecurity.
5. What is phishing attack?

CO3 (5)

1. A password “abc123” is used for an account; identify the issue in this password.
2. While using a shared computer, state one action to protect your account after use.
3. The same password is used across several websites; identify the vulnerability.
4. A login page appears similar to a bank website but has a different URL; identify the threat.
5. A user shares their password with a friend to access an account; identify the security issue.

7-Mark Questions

CO1 (5)

1. Define password manager features.
2. State the principles of a trusted system.
3. List the types of multi-factor authentication (MFA).
4. Name the data access control models.
5. Define a reference monitor.

CO2 (5)

1. Explain the importance of password complexity in cybersecurity.
2. Differentiate between Capability-based access control and Access Control Lists (ACL).
3. Explain the working of a reference monitor in system security.
4. Discuss the role of trusted systems in ensuring secure operations.
5. Explain how a password manager enhances security.

CO3 (5)

1. A user struggles to manage multiple passwords securely. How KeePassXC can be used to store, organize, and protect credentials, and examine its benefits over manual password management.
2. A web application is experiencing unauthorized login attempts. Design a secure login system and examine how each security feature helps prevent such attacks.
3. An organization plans to strengthen its authentication process. Describe how to implement a Multi-Factor Authentication (MFA) system and analyze its effectiveness against common threats.
4. A company is facing repeated password-related breaches. Propose measures to improve password security and analyze how these measures reduce security risks.
5. A system is vulnerable to boot-level attacks. Explain how to implement a trusted boot process and analyze how it ensures system integrity during startup.

10 Mark Questions

Application-Based

1. Can you design a secure password management system for employees in a mid-sized IT company? Include both technical and user education components.
2. How would you improve an existing access control system in a hospital to comply with HIPAA standards? Justify your plan using examples of trusted systems.